# A Technical Introduction to Support vector machine

## Section 1

where x i {\displaystyle x_{i}} is a training sample with target value y i {\displaystyle y_{i}} . The inner product plus intercept ■ w , x i ■ + b {\displaystyle \langle w,x_{i}\rangle +b} is the prediction for that sample, and ε {\displaystyle \varepsilon } is a free parameter that serves as a threshold: all predictions have to be within an ε {\displaystyle \varepsilon } range of the true predictions. Slack variables are usually added into the above to allow for errors and to allow approximation in the case the above problem is infeasible.

## Section 2

where the y i {\displaystyle y_{i}} are either 1 or −1, each indicating the class to which the point x i {\displaystyle \mathbf {x} _{i}} belongs. Each x i {\displaystyle \mathbf {x} _{i}} is a p {\displaystyle p} -dimensional real vector. We want to find the "maximum-margin hyperplane" that divides the group of points x i {\displaystyle \mathbf {x} _{i}} for which y i = 1 {\displaystyle y_{i}=1} from the group of points for which y i = − 1 {\displaystyle y_{i}=-1} , which is defined so that the distance between the hyperplane and the nearest point x i {\displaystyle \mathbf {x} _{i}} from either group is maximized. Any hyperplane can be written as the set of points x {\displaystyle \mathbf {x} } satisfying

## Section 3

We focus on the soft-margin classifier since, as noted above, choosing a sufficiently small value for λ {\displaystyle \lambda } yields the hard-margin classifier for linearly classifiable input data. The classical approach, which involves reducing (2) to a quadratic programming problem, is detailed below. Then, more recent approaches such as sub-gradient descent and coordinate descent will be discussed.

## Section 4

SVMs are helpful in text and hypertext categorization, as their application can significantly reduce the need for labeled training instances in both the standard inductive and transductive settings. Some methods for shallow semantic parsing are based on support vector machines. Classification of images can also be performed using SVMs. Experimental results show that SVMs achieve significantly higher search accuracy than traditional query refinement schemes after just three to four rounds of relevance feedback. This is also true for image segmentation systems, including those using a modified version SVM that uses the privileged approach as suggested by Vapnik. Classification of satellite data like SAR data using supervised SVM. Hand-written characters can be recognized using SVM. The SVM algorithm has been widely applied in the biological and other sciences. They have been used to classify proteins with up to 90% of the compounds classified correctly. Permutation tests based on SVM weights have been suggested as a mechanism for interpretation of SVM models. Support vector machine weights have also been used to interpret SVM models in the past. Posthoc interpretation of support vector machine models in order to identify features used by the model to make predictions is a relatively new area of research with special significance in the biological sciences.

## Section 5

minimize w , b , ζ ■ w ■ 2 2 + C ∑ i = 1 n ζ i subject to y i ( w ■ x i − b ) ≥ 1 − ζ i , ζ i ≥ 0 ∀ i ∈ { 1 , … , n } {\displaystyle {\begin{aligned}&{\underset {\mathbf {w} ,\;b,\;\mathbf {\zeta } }{\operatorname {minimize} }}&&\|\mathbf {w} \|_{2}^{2}+C\sum _{i=1}^{n}\zeta _{i}\\&{\text{subject to}}&&y_{i}(\mathbf {w} ^{\top }\mathbf {x} _{i}-b)\geq 1-\zeta _{i},\quad \zeta _{i}\geq 0\quad \forall i\in \{1,\dots ,n\}\end{aligned}}}

## Section 6

Note that f {\displaystyle f} is a convex function of w {\displaystyle \mathbf {w} } and b {\displaystyle b} . As such, traditional gradient descent (or SGD) methods can be adapted, where instead of taking a step in the direction of the function's gradient, a step is taken in the direction of a vector selected from the function's sub-gradient. This approach has the advantage that, for certain implementations, the number of iterations does not scale with n {\displaystyle n} , the number of data points.

## Section 7

Properties SVMs belong to a family of generalized linear classifiers and can be interpreted as an extension of the perceptron. They can also be considered a special case of Tikhonov regularization. A special property is that they simultaneously minimize the empirical classification error and maximize the geometric margin; hence they are also known as maximum margin classifiers. A comparison of the SVM to other classifiers has been made by Meyer, Leisch and Hornik.

## Section 8

Note that y i {\displaystyle y_{i}} is the i-th target (i.e., in this case, 1 or −1), and w T x i − b {\displaystyle \mathbf {w} ^{\mathsf {T}}\mathbf {x} _{i}-b} is the i-th output. This function is zero if the constraint in (1) is satisfied, in other words, if x i {\displaystyle \mathbf {x} _{i}} lies on the correct side of the margin. For data on the wrong side of the margin, the function's value is proportional to the distance from the margin. The goal of the optimization then is to minimize:

## Section 9

minimize w , b 1 2 ■ w ■ 2 subject to y i ( w ■ x i − b ) ≥ 1 ∀ i ∈ { 1 , … , n } {\displaystyle {\begin{aligned}&{\underset {\mathbf {w} ,\;b}{\operatorname {minimize} }}&&{\frac {1}{2}}\|\mathbf {w} \|^{2}\\&{\text{subject to}}&&y_{i}(\mathbf {w} ^{\top }\mathbf {x} _{i}-b)\geq 1\quad \forall i\in \{1,\dots ,n\}\end{aligned}}}

## Section 10

y i ( w T x i − b ) = 1 ■ b = w T x i − y i . {\displaystyle y_{i}(\mathbf {w} ^{\mathsf {T}}\mathbf {x} _{i}-b)=1\iff b=\mathbf {w} ^{\mathsf {T}}\mathbf {x} _{i}-y_{i}.}

## Section 11

History The original SVM algorithm was invented by Vladimir N. Vapnik and Alexey Ya. Chervonenkis in 1964. In 1992, Bernhard Boser, Isabelle Guyon and Vladimir Vapnik suggested a way to create nonlinear classifiers by applying the kernel trick to maximum-margin hyperplanes. The "soft margin" incarnation, as is commonly used in software packages, was proposed by Corinna Cortes and Vapnik in 1993 and published in 1995.

## Section 12

Polynomial (homogeneous): k ( x i , x j ) = ( x i ⋅ x j ) d {\displaystyle k(\mathbf {x} _{i},\mathbf {x} _{j})=(\mathbf {x} _{i}\cdot \mathbf {x} _{j})^{d}} . Particularly, when d = 1 {\displaystyle d=1} , this becomes the linear kernel. Polynomial (inhomogeneous): k ( x i , x j ) = ( x i ⋅ x j + r ) d {\displaystyle k(\mathbf {x} _{i},\mathbf {x} _{j})=(\mathbf {x} _{i}\cdot \mathbf {x} _{j}+r)^{d}} . Gaussian radial basis function: k ( x i , x j ) = exp ■ ( − γ ■ x i − x j ■ 2 ) {\displaystyle k(\mathbf {x} _{i},\mathbf {x} _{j})=\exp \left(-\gamma \left\|\mathbf {x} _{i}-\mathbf {x} _{j}\right\|^{2}\right)} for γ > 0 {\displaystyle \gamma >0} . Sometimes parametrized using γ = 1 / ( 2 σ 2 ) {\displaystyle \gamma =1/(2\sigma ^{2})} . Sigmoid function (Hyperbolic tangent): k ( x i , x j ) = tanh ■ ( κ x i ⋅ x j + c ) {\displaystyle k(\mathbf {x_{i}} ,\mathbf {x_{j}} )=\tanh(\kappa \mathbf {x} _{i}\cdot \mathbf {x} _{j}+c)} for some (not every) κ > 0 {\displaystyle \kappa >0} and c < 0 {\displaystyle c<0} . The kernel is related to the transform φ ( x i ) {\displaystyle \varphi (\mathbf {x} _{i})} by the equation k ( x i , x j ) = φ ( x i ) ⋅ φ ( x j ) {\displaystyle k(\mathbf {x} _{i},\mathbf {x} _{j})=\varphi (\mathbf {x} _{i})\cdot \varphi (\mathbf {x} _{j})} . The value w is also in the transformed space, with w = ∑ i α i y i φ ( x i ) {\textstyle \mathbf {w} =\sum _{i}\alpha _{i}y_{i}\varphi (\mathbf {x} _{i})} . Dot products with w for classification can again be computed by the kernel trick, i.e. w ⋅ φ ( x ) = ∑ i α i y i k ( x i , x ) {\textstyle \mathbf {w} \cdot \varphi (\mathbf {x} )=\sum _{i}\alpha _{i}y_{i}k(\mathbf {x} _{i},\mathbf {x} )} .